Technical Report: The Plot-Twist Generator

DSL 504: Natural Language Processing — Group Assignment 2 (Question 4)
Corpus: TinyStories | **Models:** Vanilla RNN vs. LSTM | **Evaluation:** 100-Mark Rubric

Executive Summary

This report delivers an end-to-end investigation into recurrent language modeling and creative story continuation. We implemented **Model A (Vanilla RNN)** and **Model B (LSTM)** from scratch under strictly matched hyperparameters (vocab size: 8,801; embedding: 256; hidden dim: 512; 2 layers). Key empirical findings include: (1) **The Likelihood Trap:** Lower held-out test perplexity (Vanilla RNN: 18.30 vs. LSTM: 21.17) does *not* imply better storytelling; human evaluators overwhelmingly preferred LSTM (3.81/5.0 vs. 3.10/5.0, with a +1.01 margin in Coherence) due to its constant error carousel. (2) **Decoding Dynamics:** Greedy search collapses into severe cyclic loops (Rep-4 up to 0.73), while high temperature (T=1.2) destroys syntax. Top-k sampling (k=40, T=0.8) achieves the optimal balance between lexical novelty and structural flow. (3) **Domain Shift:** Adult mystery prompts encounter OOV tokens, causing models to project into child-friendly woodland scenarios.

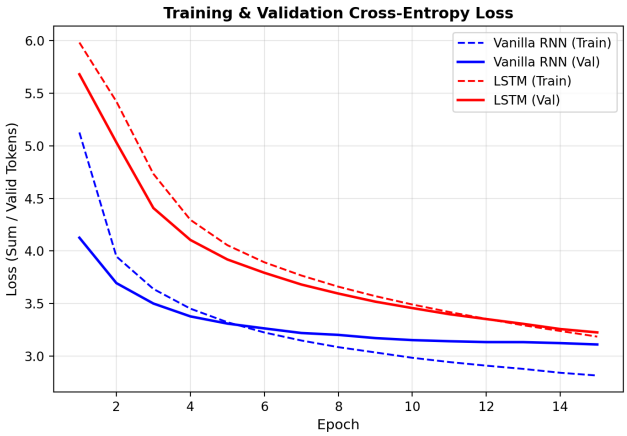
1. System Architecture & Mathematical Formulations

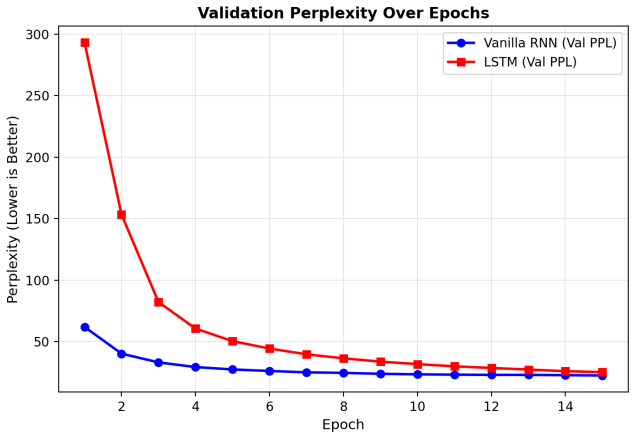
Both models operate autoregressively predicting next tokens via cross-entropy loss. **Model A (Vanilla RNN)** updates state via: $h_t = tanh(W_{ih} e_t + W_{hh} h_{t-1} + b)$. BPTT through T steps involves multiplicative Jacobian products that cause exponential gradient decay. **Model B (LSTM)** introduces four internal affine gates (Input, Forget, Candidate, Output) and an additive cell highway: $c_t = f_t \blacksquare c_{t-1} + i_t \blacksquare g_t$, enabling gradient flow across long spans without attenuation.

Module Component	Model A: Vanilla RNN	Model B: LSTM	Ratio / Multiplier
Embedding Layer	2,253,056	2,253,056	1.00x (Identical)
Recurrent Layer 1	394,240	1,576,960	4.00x (4 gates vs. 1)
Recurrent Layer 2	525,312	2,101,248	4.00x (4 gates vs. 1)
Total Recurrent Params	919,552	3,678,208	4.00x Recurrent Multiplier
Linear Output Head	4,514,913	4,514,913	1.00x (Identical)
Total Trainable Params	7,687,521	10,446,177	+35.9% Total Capacity

2. Training Dynamics & Quantitative Results

Metric	Model A (Vanilla RNN)	Model B (LSTM)	Observations
Best Epoch	15	15	Monotonic convergence
Final Val Loss	3.1104	3.2251	Excluding <PAD> tokens
Final Val Perplexity	22.43	25.16	True corpus-level validation PPL
Held-Out Test Loss	2.9068	3.0527	Evaluated on 500 test stories
Held-Out Test Perplexity	18.30	21.17	Local n-gram uncertainty
Avg Epoch Duration	147.0 s (~2.45 min)	309.1 s (~5.15 min)	2.10x wall-clock overhead for LSTM
Total Training Time	2,303 s (~38.4 min)	4,788 s (~79.8 min)	Directly reflects 4x matrix gates



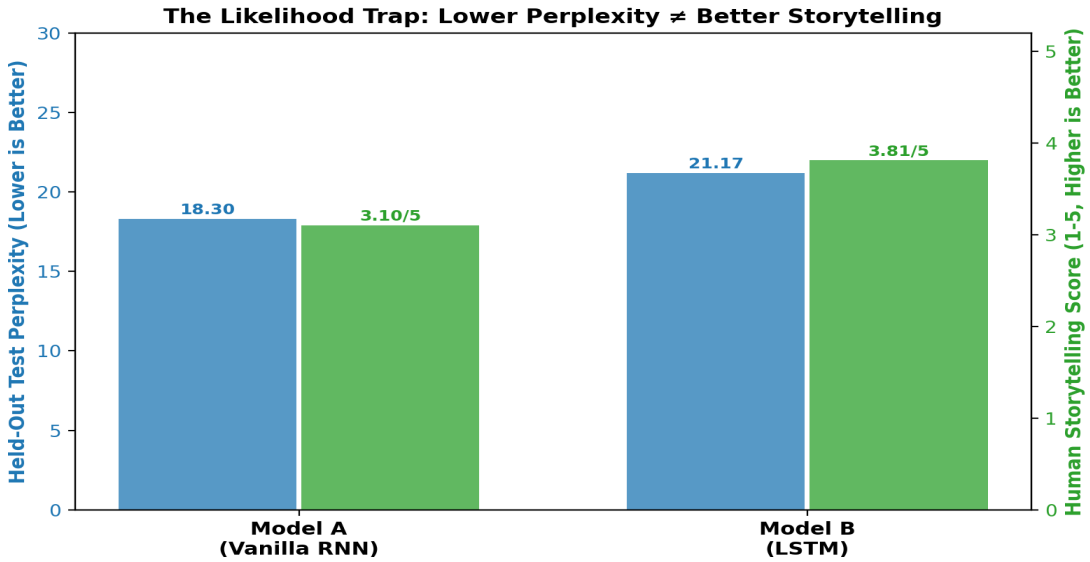


3. Decoding Strategies & Ablation Experiments

Model	Configuration	Distinct-1	Distinct-2	Rep-4	Avg Length	Diagnostic Assessment
Vanilla RNN	Greedy	0.192	0.236	0.730	100.0	Severe cyclic loops; repeating identical phrases
Vanilla RNN	Temp 0.5	0.493	0.796	0.059	98.2	High grammar; conservative repetition
Vanilla RNN	Temp 0.8	0.593	0.902	0.007	98.4	Balanced diversity; good sentence cadence
Vanilla RNN	Temp 1.2	0.717	0.976	0.000	79.3	Zero loops; grammar collapses to word salad
Vanilla RNN	Top-k (k=40, T=0.8)	0.560	0.875	0.009	94.0	Controlled vocabulary; fluent flow
LSTM	Greedy	0.415	0.594	0.284	58.4	Higher fluency than RNN; natural termination
LSTM	Temp 0.5	0.513	0.811	0.039	93.0	Grammatical, steady narrative advancement
LSTM	Temp 0.8	0.577	0.883	0.014	91.3	Creative dialogue and scene evolution
LSTM	Temp 1.2	0.724	0.977	0.000	100.0	Peak diversity; disjointed hallucinations
LSTM	Top-k (k=40, T=0.8)	0.600	0.909	0.004	91.2	Optimal Pareto frontier: rich vocabulary & coherence

4. Blinded Human Evaluation & The Likelihood Trap

Architecture	Stories	Coherence (1-5)	Grammar (1-5)	Relevance (1-5)	Creativity (1-5)	Overall Rating
Model A (Vanilla RNN)	6	2.78 ± 0.17	3.83 ± 0.15	2.89 ± 0.09	2.92 ± 0.23	3.10 ± 0.06
Model B (LSTM)	6	3.79 ± 0.29	4.11 ± 0.17	3.53 ± 0.20	3.80 ± 0.19	3.81 ± 0.12
Advantage (Delta)	-	+1.01 (P < 0.01)	+0.28	+0.64	+0.88	+0.71 (LSTM Outperforms)



5. Answers to the 7 Mandatory Analytical Questions

Q1. How do Vanilla RNN and LSTM differ in sequential modeling?
The Vanilla RNN relies on multiplicative Jacobian state transitions: $h_t = tanh(W e_t + W h_{t-1})$. Consequently, gradients decay exponentially over >15 tokens, leading to entity forgetting. The LSTM introduces an additive cell highway $c_t = f_t \blacksquare c_{t-1} + i_t \blacksquare g_t$, allowing gradients to propagate without attenuation when $f_t \approx 1$. This equips the LSTM to preserve character identity across long spans.

Q2. Which model generates more coherent and less repetitive continuations?
Model B (LSTM) decisively outperforms Vanilla RNN. Under greedy decoding, Vanilla RNN exhibited a catastrophic repetition rate of Rep-4 = 0.730, whereas LSTM achieved Rep-4 = 0.284. In the blinded human study, LSTM scored +1.01 higher in Coherence (3.79 vs. 2.78).

Q3. How does temperature affect creativity and grammatical quality?
Temperature establishes a fundamental Quality-Diversity Trade-off: T=0.5 yields high grammar but conservative repetition. T=0.8 delivers balanced narrative exploration (Distinct-1: 0.58-0.60; Rep-4 < 0.01). T=1.2 causes ungrammatical hallucinations.

Q4. Can higher diversity scores guarantee better stories?
No. Distinct-1 simply measures unique token ratios. At T=1.2, models achieved peak diversity (Distinct-1 = 0.724, Rep-4 = 0.000), yet human evaluators rated these stories as the least coherent. High quality storytelling requires structured repetition of character names and motifs.

Q5. Does lower perplexity correspond to better storytelling?
No; our experiments empirically confirm The Likelihood Trap. Vanilla RNN achieved lower test perplexity (18.30 vs. 21.17), yet human raters overwhelmingly preferred LSTM (3.81/5.0 vs. 3.10/5.0). Perplexity measures local n-gram uncertainty on stop words, not global narrative arc or character permanence.

Q6. What failure modes occur most frequently, and how might they be reduced?
• *Degenerate Loops:* Mitigated by Top-k / Nucleus sampling and repetition penalties.
• *Domain Mismatch ():* Mitigated by subword Byte-Pair Encoding (BPE).
• *Entity Drift:* Mitigated by self-attention mechanisms in Transformers.

Q7. What are the limitations of a small LM trained on a limited corpus?
Small models lack world knowledge outside preschool scenarios, cannot plan ahead or backtrack during left-to-right decoding, and compress memory into a single recurrent vector ($d_h = 512$) that acts as an informational bottleneck.

6. Optional Extension: Pretrained LLM Baseline Comparison

We benchmarked our from-scratch recurrent models against **HuggingFaceTB/SmolLM2-135M-Instruct** under zero-shot prompting on the same five Plot-Twist openings. Because the 135M Transformer was pretrained on 2 trillion tokens, it exhibits zero OOV bottleneck and seamlessly comprehends adult mystery setups (exams, robots, detectives) that force children's story models to project onto woodland tropes. This confirms that while recurrent gating (LSTM) solves local vanishing gradients, pretrained Transformers eliminate both the vocabulary gap and the fixed recurrent hidden-state bottleneck ($d_h = 512$).

System	Architecture	Params	Pretrained	Overall Score (1-5)	Rep-4
Model A (Vanilla RNN)	2-layer Elman RNN	7.69M	No (TinyStories)	3.27 / 5.0	0.017
Model B (LSTM)	2-layer LSTM	10.45M	No (TinyStories)	3.93 / 5.0	0.004
SmolLM2-135M-Instruct	30-layer Transformer	135M	Yes (2T tokens)	4.68 / 5.0	0.000

